

Memory Learning Mode

1. Function Description

Achieve action learning and execution through voice commands.

2. Startup and Operation

2.1 Startup

Open the terminal and enter the following command to start:

```
python3 ~/dofbot_voice/scripts/learnning_model.py
```

2.2 Operation Steps

After the program runs successfully, the voice module will broadcast "I'm here", then say "Hi, yahboom" to the voice module. The voice module replies "I'm here" indicating successful wake-up.

2.2.1 Learning Actions

Move the robotic arm to the desired pose, then say "Record it" to the voice module. If fewer than 20 groups of actions have been input, the voice module will reply "OK". If 20 groups of actions have already been input, the voice module will reply "Too many actions, I Can't record them". If you want to end the input, say "Over" to the voice module, and the voice module will reply "ok".

2.2.2 Playing Actions

Say "Display actions" to the voice module, the voice module will reply "ok", and the robotic arm will move to the learned poses sequentially.

2.2.3 Clearing Actions

Say "Clear actions" to the voice module, the voice module will reply "ok", then you can proceed to the next learning session.

3. Core Code Analysis

Source code path: `~/dofbot_voice/scripts/learnning_model.py`

```
# Call function to get voice recognition result
result = mySpeech.speech_read()
# If the voice recognition result is 55, it means learning and recording the
current robotic arm pose is required
if result == 55:
    time.sleep(.1)
    Arm.Arm_Action_Study()
    time.sleep(.5)
    num = Arm.Arm_Read_Action_Num()
    print(num)
    time.sleep(.5)
    # If less than 20 groups of actions, broadcast "OK, please input the next
    group of actions"
```

```
if num < 20:
    mySpeech.void_write(55)
    # If greater than or equal to 20 groups of actions, broadcast "More than 20
    groups of actions, recording ended"
elif num >= 20:
    mySpeech.void_write(56)
    # If the voice recognition result is 56, it means ending the action learning
    recording
elif result == 56:
    mySpeech.void_write(57)
    Arm.Arm_Button_Mode(0)
    # If the voice recognition result is 57, it means playing the learned actions
elif result == 57:
    mySpeech.void_write(58)
    Arm.Arm_Action_Mode(1)
    # If the voice recognition result is 58, it means clearing actions in the action
    list
elif result == 58:
    mySpeech.void_write(59)
    Arm.Arm_Clear_Action()
```